

SPRINT 5: BASES DE DATOS NO RELACIONALES

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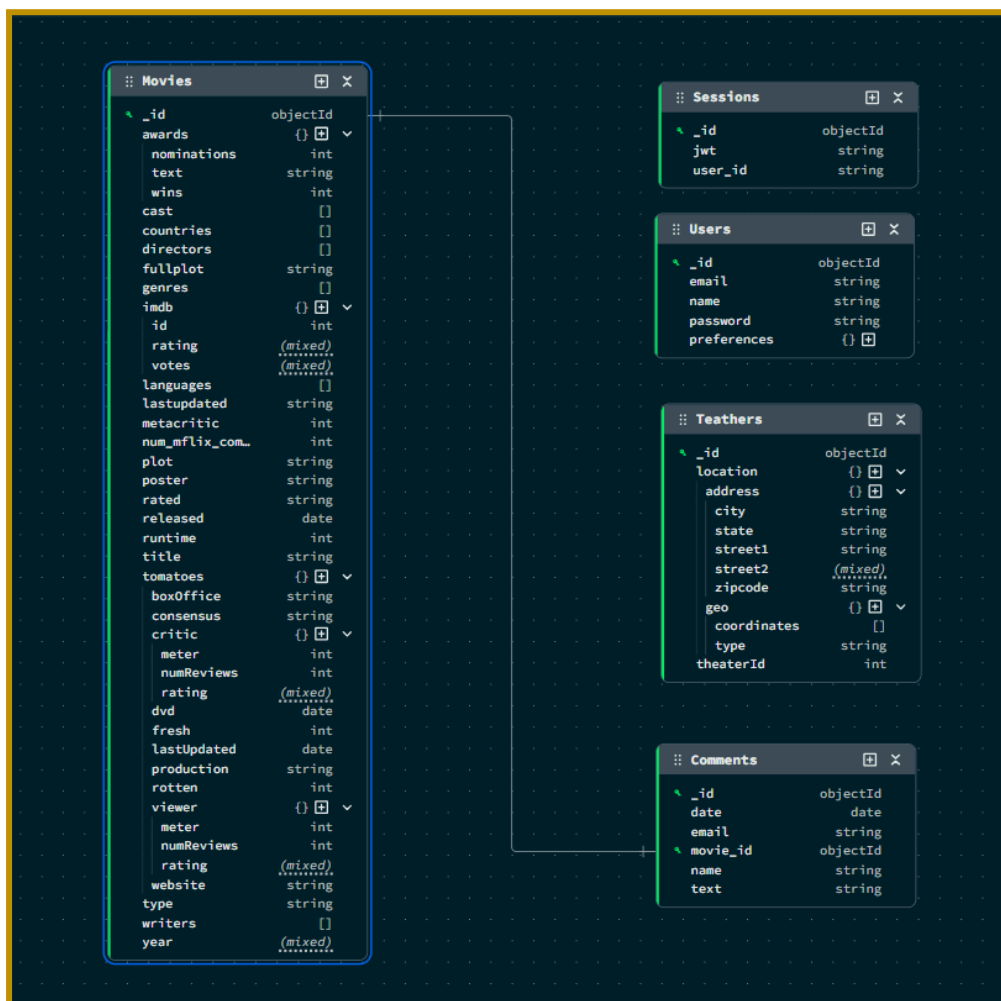
NIVEL 1

EJERCICIO 1

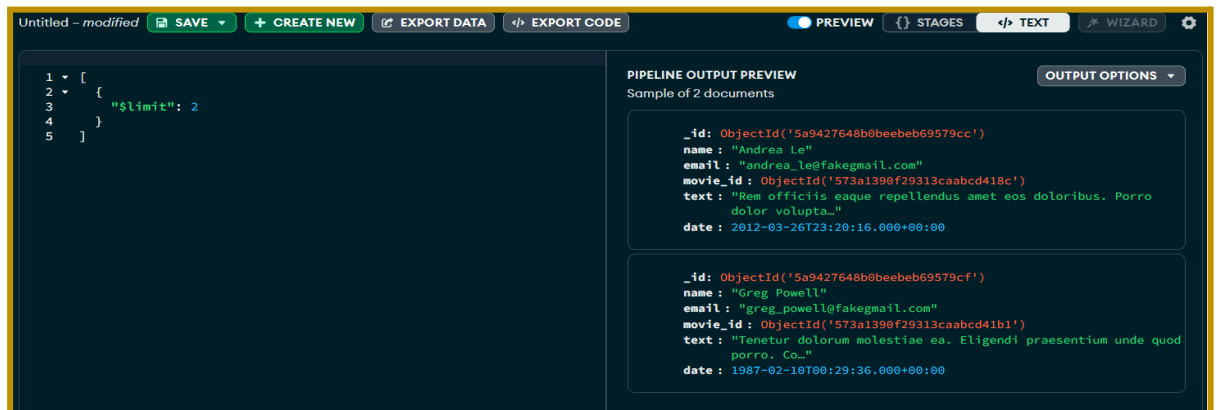
Crear la base de datos:



Collection name	Properties	Storage size	Data size	Documents	Avg. document size	Indexes	Total index size
Sessions	-	4.10 kB	540.00 B	1	540.00 B	1	4.10 kB
Users	-	4.10 kB	29.57 kB	185	159.00 B	1	4.10 kB
Teathers	-	4.10 kB	349.83 kB	1.6K	223.00 B	1	4.10 kB
Movies	-	4.10 kB	37.62 MB	24K	1.60 kB	1	4.10 kB
Comments	-	4.10 kB	14.29 MB	50K	284.00 B	1	4.10 kB



- Muestra los 2 primeros comentarios que aparecen en la base de datos.



```
1 [
2   {
3     "$limit": 2
4   }
5 ]
```

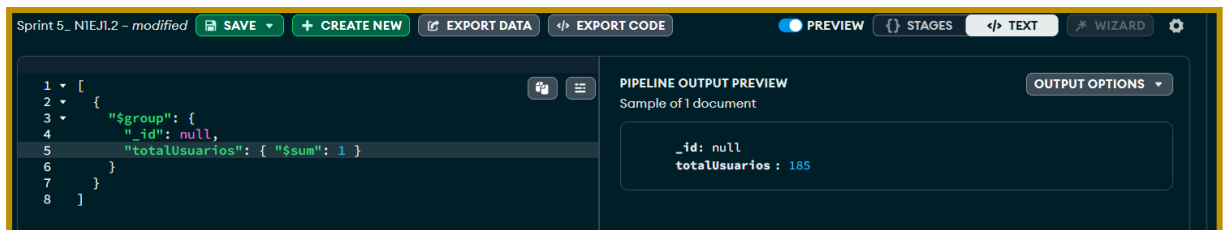
PIPELINE OUTPUT PREVIEW
Sample of 2 documents

OUTPUT OPTIONS ▾

```
_id: ObjectId('5a9427648b0beebe69579cc')
name: "Andrea Le"
email: "andrea_le@fakegmail.com"
movie_id: ObjectId('573a1398f29313caabcd418c')
text: "Rem officis eaque repellendus amet eos doloribus. Porro
      dolor volupta..."
date: 2012-03-26T23:20:16.000+00:00
```

```
_id: ObjectId('5a9427648b0beebe69579cf')
name: "Greg Powell"
email: "greg_powell@fakegmail.com"
movie_id: ObjectId('573a1398f29313caabcd41b1')
text: "Tenetur dolorum molestiae ea. Eligendi praesentium unde quod
      porro. Co..."
date: 1987-02-10T00:29:36.000+00:00
```

- ¿Cuántos usuarios tenemos registrados?



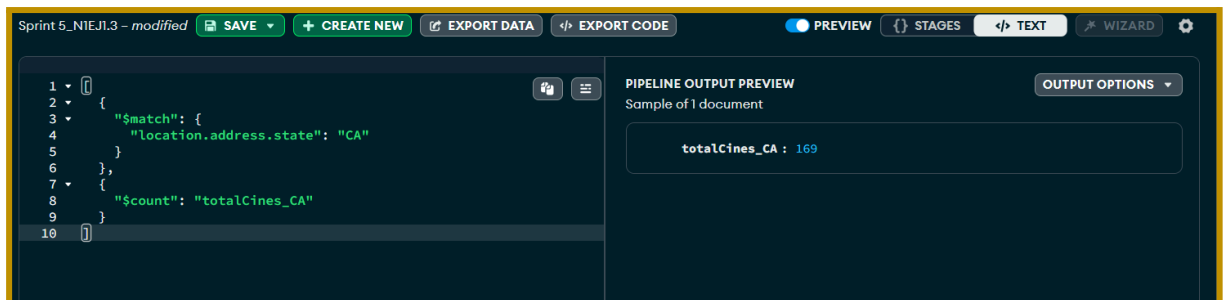
```
1 [
2   {
3     "$group": {
4       "_id": null,
5       "totalUsuarios": { "$sum": 1 }
6     }
7   }
8 ]
```

PIPELINE OUTPUT PREVIEW
Sample of 1 document

OUTPUT OPTIONS ▾

```
_id: null
totalUsuarios: 185
```

- ¿Cuántos cines existen en el estado de California?



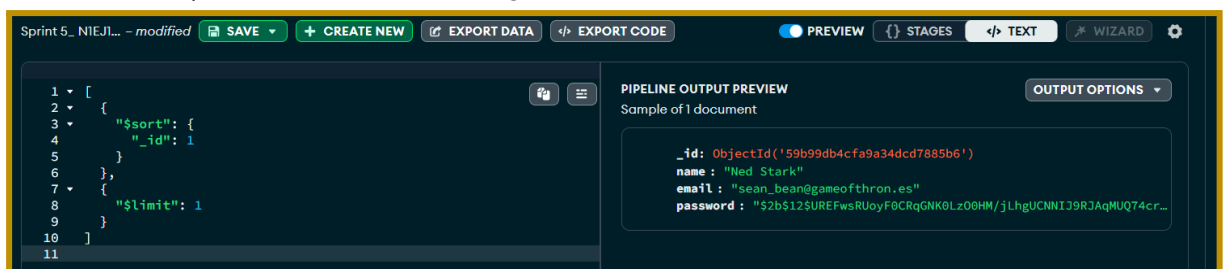
```
1 [
2   {
3     "$match": {
4       "location.address.state": "CA"
5     }
6   },
7   {
8     "$count": "totalCines_CA"
9   }
10 ]
```

PIPELINE OUTPUT PREVIEW
Sample of 1 document

OUTPUT OPTIONS ▾

```
totalCines_CA: 169
```

- ¿Cuál fue el primer usuario en registrarse?



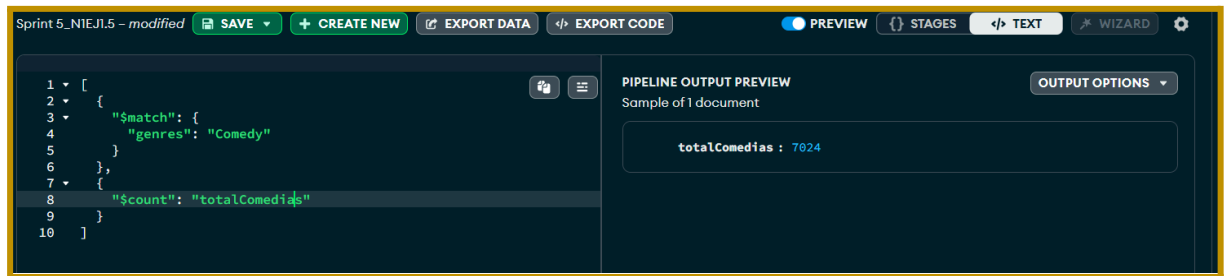
```
1 [
2   {
3     "$sort": {
4       "_id": 1
5     }
6   },
7   {
8     "$limit": 1
9   }
10 ]
11
```

PIPELINE OUTPUT PREVIEW
Sample of 1 document

OUTPUT OPTIONS ▾

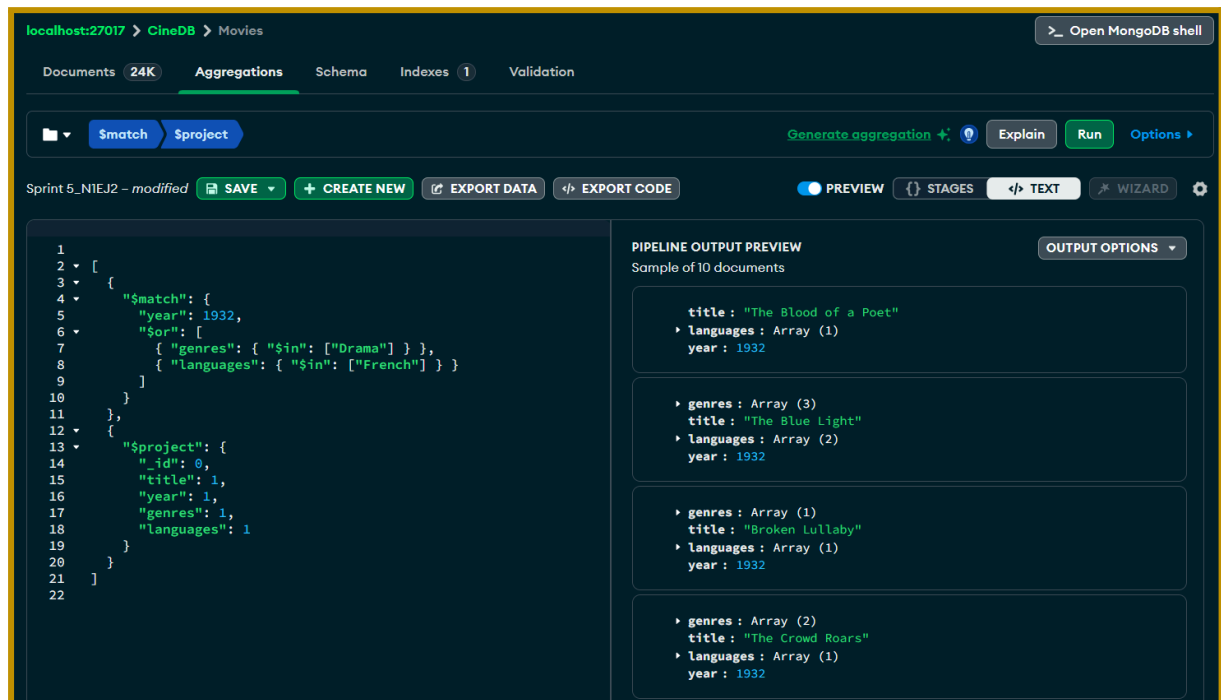
```
_id: ObjectId('59b99db4cfa9a34dcd7885b6')
name: "Ned Stark"
email: "sean_bean@gameofthron.es"
password: "$2b$12$UREFwsRUoyF0CrqGNK0Lz00HM/jLhgUCNNI39RJAqMUQ74cr...
```

- ¿Cuántas películas de comedia existen en nuestra base de datos?



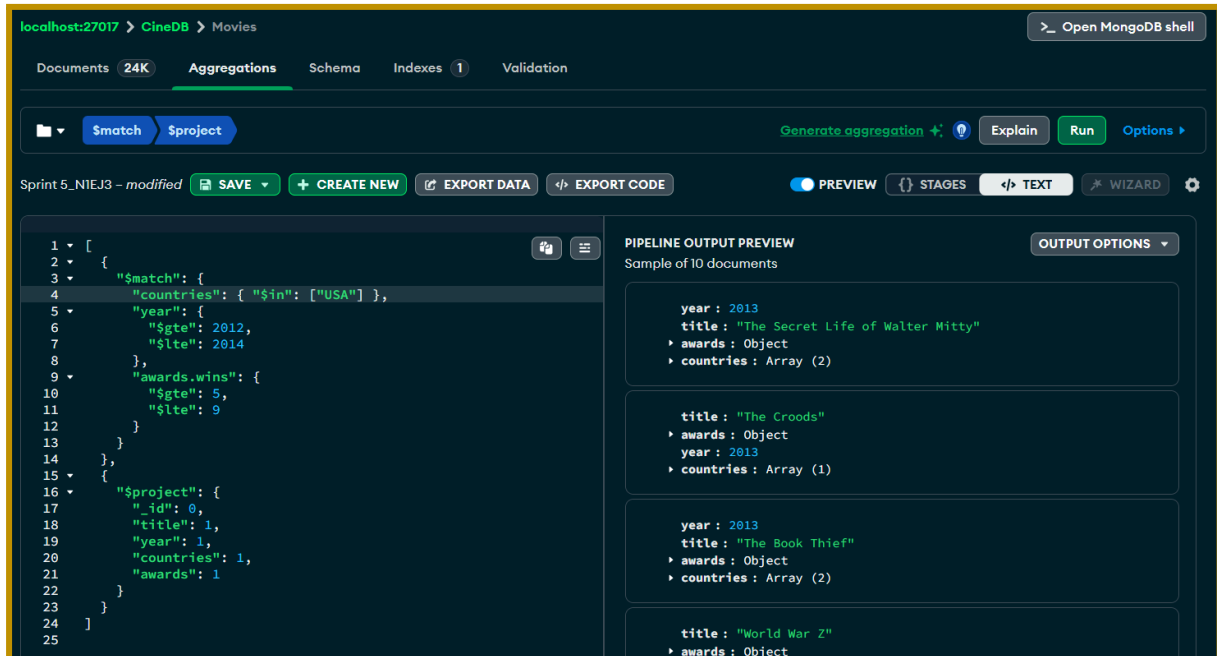
EJERCICIO 2

Muéstrame todos los documentos de las películas producidas en 1932, pero que el género sea drama o estén en francés



EJERCICIO 3

Muéstrame todos los documentos de películas estadounidenses que tengan entre 5 y 9 premios que fueron producidas entre 2012 y 2014.



The screenshot shows the MongoDB Compass interface for the 'CineDB' database, specifically the 'Movies' collection. The 'Aggregations' tab is active, showing a pipeline with two stages: '\$match' and '\$project'. The '\$match' stage filters for documents where 'countries' is 'USA', 'year' is between 2012 and 2014, and 'awards.wins' is between 5 and 9. The '\$project' stage projects the '_id', 'title', 'year', 'countries', and 'awards'. The 'PIPELINE OUTPUT PREVIEW' on the right shows a sample of 10 documents, including 'The Secret Life of Walter Mitty', 'The Croods', 'The Book Thief', and 'World War Z'.

```
1 [
2   {
3     "$match": {
4       "countries": { "$in": ["USA"] },
5       "year": {
6         "$gte": 2012,
7         "$lte": 2014
8       },
9       "awards.wins": {
10        "$gte": 5,
11        "$lte": 9
12      }
13    },
14    "$project": {
15      "_id": 0,
16      "title": 1,
17      "year": 1,
18      "countries": 1,
19      "awards": 1
20    }
21  }
22 ]
```

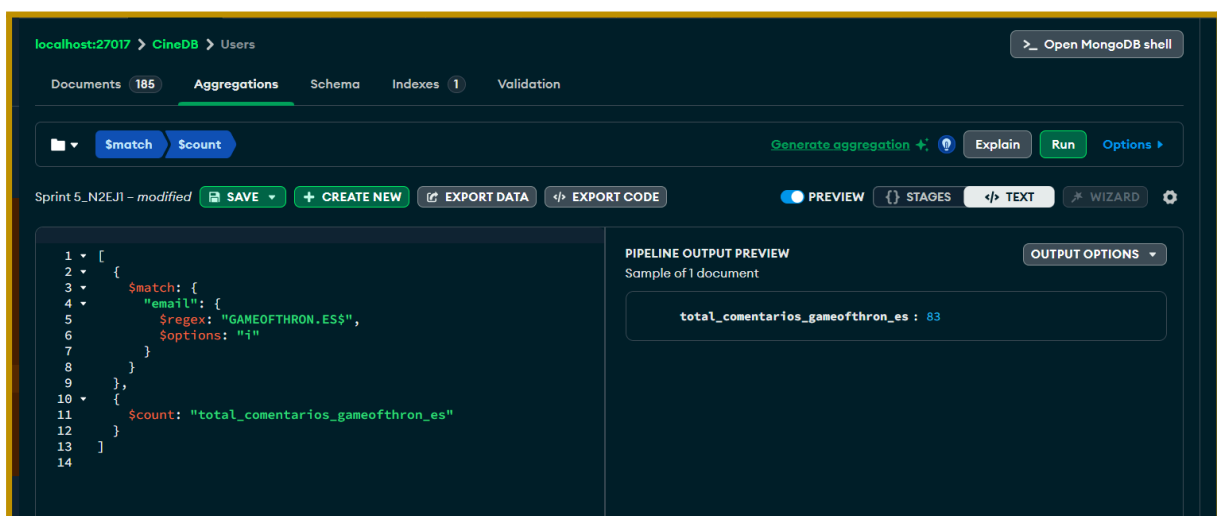
PIPELINE OUTPUT PREVIEW
Sample of 10 documents

- year: 2013
title: "The Secret Life of Walter Mitty"
awards: Object
countries: Array (2)
- title: "The Croods"
awards: Object
year: 2013
countries: Array (1)
- year: 2013
title: "The Book Thief"
awards: Object
countries: Array (2)
- title: "World War Z"
awards: Object

NIVEL 2

EJERCICIO 1

Cuenta cuántos comentarios escribe un usuario que utiliza "GAMEOFTHRON.ES" como dominio de correo electrónico.



The screenshot shows the MongoDB Compass interface for the 'CineDB' database, specifically the 'Users' collection. The 'Aggregations' tab is active, showing a pipeline with two stages: '\$match' and '\$count'. The '\$match' stage filters for documents where 'email' contains 'GAMEOFTHRON.ES'. The '\$count' stage counts the total number of documents for each user. The 'PIPELINE OUTPUT PREVIEW' on the right shows a sample of 1 document with the result: 'total_comentarios_gameofthron_es: 83'.

```
1 [
2   {
3     "$match": {
4       "email": {
5         "$regex": "GAMEOFTHRON.ES$",
6         "$options": "i"
7       }
8     },
9     "$count": "total_comentarios_gameofthron_es"
10  }
11 ]
```

PIPELINE OUTPUT PREVIEW
Sample of 1 document

total_comentarios_gameofthron_es: 83

EJERCICIO 2

¿Cuántos cines existen en cada código postal situados dentro del estado Washington DC (DC)?

The screenshot shows the MongoDB Compass interface for the 'CineDB' database. The 'Aggregations' tab is active, displaying a pipeline with three stages: \$match, \$group, and \$sort. The \$match stage filters for 'location.address.state': 'DC'. The \$group stage groups by 'location.address.zipcode' and counts the number of theaters ('totalCines'). The \$sort stage sorts by 'totalCines' in descending order. The 'PIPELINE OUTPUT PREVIEW' shows a sample of 3 documents with their _id and totalCines values.

```
1 [
2   {
3     $match: {
4       "location.address.state": "DC"
5     },
6   },
7   {
8     $group: {
9       _id: "$location.address.zipcode",
10      totalCines: { $sum: 1 }
11    },
12  },
13  {
14    $sort: { totalCines: -1 }
15  }
16 ]
```

PIPELINE OUTPUT PREVIEW
Sample of 3 documents

_id	totalCines
"20016"	1
"20010"	1
"20002"	1

NIVEL 3

EJERCICIO 1

Encuentra todas las películas dirigidas por John Landis con una puntuación IMDb (Internet Movie Database) de entre 7,5 y 8.

The screenshot shows the MongoDB Compass interface for the 'CineDB' database. The 'Aggregations' tab is active, displaying a pipeline with two stages: \$match and \$project. The \$match stage filters for 'directors': 'John Landis' and 'imdb.rating' between 7.5 and 8. The \$project stage projects specific fields: title, year, genres, directors, runtime, and imdb.rating. The 'PIPELINE OUTPUT PREVIEW' shows a sample of 4 documents with their _id and the projected fields.

```
1 [
2   {
3     $match: {
4       directors: "John Landis",
5       "imdb.rating": {
6         $gte: 7.5,
7         $lte: 8
8       }
9     },
10  },
11  {
12    $project: {
13      title: 1,
14      year: 1,
15      genres: 1,
16      runtime: 1,
17      directors: 1,
18      "imdb.rating": 1,
19      _id: 0
20    }
21  }
22 ]
```

PIPELINE OUTPUT PREVIEW
Sample of 4 documents

_id	title	year	genres	runtime	directors	imdb.rating
"Animal House"	"Animal House"	1978	Array (1)	109	Array (1)	7.5
"The Blues Brothers"	"The Blues Brothers"	1980	Array (3)	133	Array (1)	7.5
"The Blues Brothers"	"The Blues Brothers"	1980	Array (3)	133	Array (1)	7.5
"The Blues Brothers"	"The Blues Brothers"	1980	Array (3)	133	Array (1)	7.5

EJERCICIO 2

Muestra en un mapa la ubicación de todos los teatros de la base de datos.

